



Loan Default Prediction

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Problem

- For financial institutions, loan defaults represent a significant financial risk. When borrowers fail to repay loans, the institution incurs losses, which can accumulate and potentially threaten the institution's financial stability.
- By accurately predicting which applicants are likely to default, banks and lending institutions can increase their profits and reduce their losses.

Solution

- Build a predictive classification model that will help the business assess the risk of new loan applicants and make informed decisions on home loan approvals.
- The model must identify importance of different features, perform well on unseen data sets, and maintain a good balance between false positives and false negatives.

Data set considered

The Home Equity dataset contains 5,960 records with the following 13 columns/features

	Column/Feature	Description	Populated values	Missing values	% Missing values
1	DEBTINC	Debt to income ratio	4693	1267	21.3%
2	DEROG	Derogatory reports	5252	708	11.9%
3	DELINQ	Delinquencies	5380	580	9.7%
4	MORTDUE	Existing mortgage	5442	518	8.7%
5	YOJ	Years at present job	5445	515	8.6%
6	NINQ	Number of credit inquiries	5450	510	8.6%
7	CLAGE	Age of oldest credit line in months	5653	307	5.2%
8	JOB	Job type	5681	279	4.7%
9	REASON	Reason for taking loan	5708	252	4.2%
10	CLNO	Number of existing credit lines	5738	222	3.7%
11	VALUE	Current value of property	5848	112	1.9%
12	LOAN	Loan amount approved	5960	0	0%
13	BAD	0 = Defaulted, 0 = Repaid	5960	0	0%

Approach

1. Data Preparation

- Clean the dataset by removing duplicates, addressing missing values, and encoding categorical variables into numeric form.

2. Model Building

- Split the data into 80% training and 20% test data. Train multiple models (Logistic Regression, Decision Tree, Random Forest, etc.) on the training data.

3. Model Evaluation and Selection

- Evaluate the model performance on the training data set. Test the models on the test data set to ensure they generalize well on new real-world data.

4. Hyperparameter Tuning

- Fine-tune the selected model's hyperparameters using grid search or random search to optimize performance.

5. Final Model Validation

- Select the model with the highest recall and precision, to minimize financial risk while ensuring that good customers are not unfairly penalized.

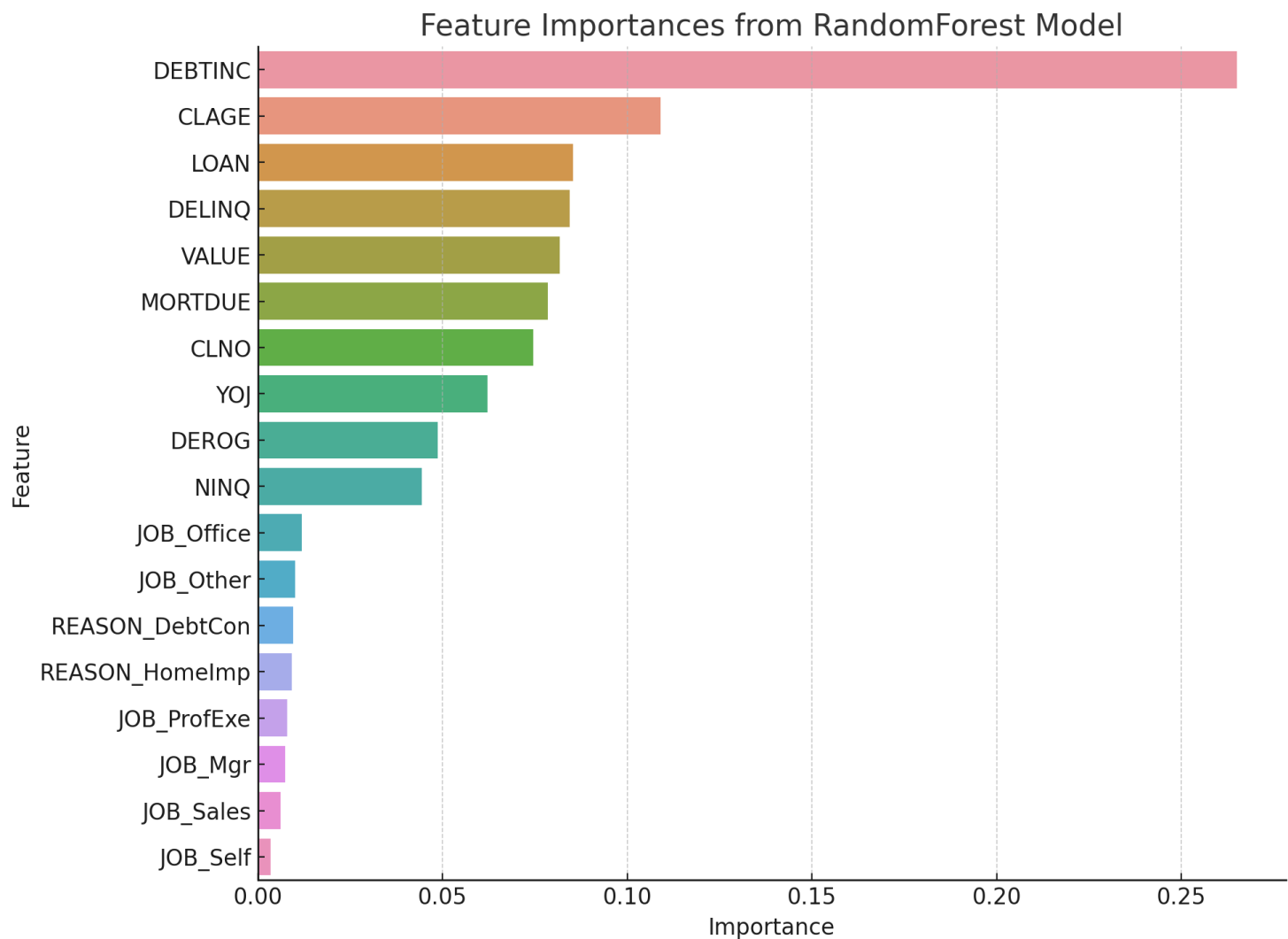
6. Model Deployment

- Deploy the model for real-time prediction. Monitor the model's performance over time. Retrain as needed to account for changes in the data distribution.



Key Findings

Relative impact of features on making accurate default predictions



Takeaways

DEBTINC (Debt to Income ratio) is the highest predictor of loan default.

CLAGE (age of the oldest credit line) is the second most important feature. A longer credit history might indicate a more established and possibly more reliable borrower.

DELINQ (delinquencies) is an important predictor.

VALUE and **MORTDUE** are also important, indicating that the equity an applicant has in their property is relevant to their likelihood of default.

REASON and **JOB** are least critical.

Test results from different models

	Model	Recall	Precision	F1 Score	Accuracy
1	Logistic Regression	0.0339623	0.818182	0.0652174	0.783557
2	Decision Tree	0.622642	0.723684	0.669371	0.863255
3	Tuned Decision Tree	0.830189	0.637681	0.721311	0.857383
4	Random Forest	0.701887	0.873239	0.778243	0.911074
5	Tuned Random Forest	0.781132	0.676471	0.725044	0.868289



Takeaways

Decision Tree catches more defaulters (higher recall).

Random Forest avoids more false alarms (higher precision).

Recommendation

Random Forest - because its higher precision will help in catching significant number of defaulters while avoiding unnecessary loan rejections (higher F1 Score).

Breakup of test results by class (BAD=0/1)

Model	Class	Recall	Precision	F1 Score	Count
Decision Tree	Class 0	0.93	0.90	0.91	927
	Class 1	0.62	0.72	0.67	265
	Weighted Avg	0.86	0.86	0.86	1192
Tuned Decision Tree	Class 0	0.87	0.95	0.90	927
	Class 1	0.83	0.64	0.72	265
	Weighted Avg	0.88	0.86	0.86	1192
Random Forest	Class 0	0.97	0.92	0.94	927
	Class 1	0.70	0.87	0.78	265
	Weighted Avg	0.91	0.91	0.91	1192
Tuned Random Forest	Class 0	0.89	0.93	0.91	927
	Class 1	0.78	0.68	0.73	265
	Weighted Avg	0.88	0.87	0.87	1192



Takeaways

Tuning the Decision Tree improved its recall from 0.62 to 0.83 for class 1.

Tuning the Random Forest did not provide any benefit over Random Forest.

Overall, Random Forest performed best.

Recommendation

- The best model to predict which customers are likely to default is the **Random Forest Classifier**.
 - This model had 87% Precision, 70% Recall, and 78% F1 Score.
 - Although Decision Tree catches more defaulters (higher recall), Random Forest avoids more false alarms (higher precision).
 - Random Forest has the highest F1 score, which means it provides a better balance because its higher precision will help in catching significant number of defaulters while avoiding unnecessary loan rejections.
- Deploy the model for real-time prediction. Monitor the model's performance over time. Retrain as needed to account for changes in the data distribution.